

tory bodies such as the Food and Drug Administration (FDA) in the US have been caught off guard by the flood of new devices, and they are now scrambling to address the issue.

Another concern relates to device longevity. The two-year life-cycle of any mobile phone model does not match well with the expected lifecycle of a portable medical device, which could be seven years or more. This raises concerns regarding component supply and product support.

Third is the emergence of low-cost communication standards—both wired and wireless. Larger medical devices are typically used in a hospital or laboratory setting, and results often centralised via computer subsystems connected to a wired Internet connection. The same need for storage of results exists with portable equipment too.

The emergence of wired communication standards such as USB and wireless standards such as Bluetooth have played an important role in easing the communication design and data interoperability challenges. In the wired domain, USB has become ubiquitous, but the wireless situation is still in a state of flux, with Bluetooth (especially Bluetooth Low Energy), Zigbee, the upcoming Medical Body Area Network (MBAN) standard and several others vying for dominance.

Components at the heart

A major trend is the development of devices that can be used by people who have no special training. The component at the centre of this trend is the microcontroller unit (MCU). Making smarter portable devices calls for more capable MCUs. The MCU applies algorithms and programmed data in order to accomplish the intended function of the device.

Besides the MCU, most portable medical-electronic devices have at least one sensor to convert temperature, pressure, sound, electrochemical reactions or some other physical quantity into an electrical signal. These signals often need some level of processing before these can be used by the MCU.



Polar FT7 heart rate monitor

This processing is done by analogue components, such as operational amplifiers, filters and analogue-to-digital converters.

MEMS is the technology behind various types of silicon sensors and nano pumps. Consumer sensor technologies from STMicroelectronics include multiple-axis motion and orientation, compass, pressure, temperature, sound (microphone) sensors as well as biological and molecular compounds.

In some portable medical-electronic systems, the MCU needs to be able to affect the outside world. This can be accomplished via motors, linear actuators, pumps, fans, lamps and other components that are connected to the MCU. Depending on the amount of power involved, some of these components may have a field-effect transistor (FET) or other power-control device between them and the MCU.

Most portable medical-electronic devices include some means of communicating with the user. This can be via lights and buttons, an LCD or touch-screen, audio cues including speech, or through a remote interface.

Practically, all of these portable products need a low-power MCU to take commands from the user or operator and provide readings and status updates. Because almost all of these products run on batteries, a key con-

sideration is lengthening the battery life. Products vary in the battery lifetime required, but two to ten years between battery changes is not unusual.

Robust data processing and a wired or wireless communication interface are important. High-

ly integrated analogue components are required to condition and convert the signals into the digital domain so that these can be processed and interpreted. Portable medical equipment can greatly benefit from the level of integration that the latest SoCs with on-board MCUs provide.

Addressing the key concerns

Semiconductor manufacturers are coming up with innovative solutions to make medical devices portable, low-power, safe, secure, reliable and connected.

Portability. The device should be easy to use wherever the patient happens to be. Here, one very important factor is the physical size of the device, which is partly determined by the size of components used inside.

Semiconductor manufacturers are working hard to reduce both the size and power requirement of portable medical-electronic devices. Consider, for example, accelerometers, which are used in pedometers and blood pressure meters. Devices such as the ADXL345 3-axis accelerometer can operate with a typical power drain of only 25 μ A in a 3 $\times$ 5 $\times$ 1mm³ package.

Standby current is especially important in many applications to conserve battery life. Devices such as the ADXL345 can run with standby power levels of 0.1 μ A, giving a couple of